

# Long Division

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

## HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

# How Many Fit

Estimate before you divide anything.

## Warm-Up 12 ITEMS • RECALL

1 How many 10s in 80

\_\_\_\_\_

2 How many 20s in 60

\_\_\_\_\_

3 How many 30s in 90

\_\_\_\_\_

4  $90 \div 30$ 

\_\_\_\_\_

5  $120 \div 40$ 

\_\_\_\_\_

6  $200 \div 50$ 

\_\_\_\_\_

7 How many 5s in 30

\_\_\_\_\_

8 How many 4s in 24

\_\_\_\_\_

9 How many 10s in 90

\_\_\_\_\_

10 How many 6s in 42

\_\_\_\_\_

11 How many 8s in 64

\_\_\_\_\_

12 How many 3s in 27

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. About how many 30s in 400 (13 thirties is 390)

.....

14.  $600 \div 20$ 

.....

15.  $840 \div 40$ 

.....

16.  $960 \div 30$ 

.....

17.  $1200 \div 60$ 

.....

18. How many 12s in 144

.....

19. How many 15s in 90

.....

20. How many 25s in 200

.....

21. How many 20s in 340

.....

22. How many 11s in 121

.....



# Long Division

Divide, multiply, subtract, bring down. Every answer here comes out whole.

## Warm-Up 12 ITEMS • RECALL

1  $84 \div 12$

2  $96 \div 16$

3  $75 \div 15$

4  $91 \div 13$

5  $144 \div 12$

6  $156 \div 13$

7  $60 \div 5$

8  $84 \div 4$

9  $96 \div 3$

10  $70 \div 5$

11  $90 \div 6$

12  $120 \div 4$

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $552 \div 24$

14.  $784 \div 28$

15.  $945 \div 35$

16.  $1,224 \div 36$

17.  $1,472 \div 32$

18.  $336 \div 6$

19.  $432 \div 8$

20.  $595 \div 7$

21.  $828 \div 9$

22.  $744 \div 6$



# Four Digits

Longer, not harder. Watch the place where the quotient has a zero.

## Warm-Up 12 ITEMS • RECALL

1  $1,000 \div 10$

2  $2,400 \div 12$

3  $3,600 \div 12$

4  $4,200 \div 20$

5  $5,000 \div 25$

6  $6,300 \div 30$

7  $1000 \div 10$

8  $2000 \div 20$

9  $3600 \div 6$

10  $4800 \div 8$

11  $1200 \div 4$

12  $5000 \div 10$

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $3,384 \div 24$

14.  $4,536 \div 21$

15.  $7,308 \div 36$  (Note the zero in the middle)

16.  $8,464 \div 46$

17.  $9,072 \div 24$

18.  $3,672 \div 12$

19.  $5,488 \div 14$

20.  $7,056 \div 24$

21.  $4,225 \div 25$

22.  $6,336 \div 16$



# Remainders

The situation decides what happens to the leftover.

## Warm-Up 11 ITEMS • RECALL

1  $17 \div 5$  — the remainder

\_\_\_\_\_

2  $17 \div 5$  — the whole-number quotient

\_\_\_\_\_

3  $23 \div 4$  — remainder

\_\_\_\_\_

4  $50 \div 7$  — remainder

\_\_\_\_\_

5  $100 \div 9$  — remainder

\_\_\_\_\_

6  $45 \div 6$  — remainder

\_\_\_\_\_

7  $13 \div 4$  — the remainder

\_\_\_\_\_

8  $20 \div 6$  — the remainder

\_\_\_\_\_

9  $25 \div 4$  — the whole part

\_\_\_\_\_

10  $31 \div 7$  — the remainder

\_\_\_\_\_

11  $19 \div 3$  — the whole part

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12. 127 children, 30 per bus. Buses needed (You cannot leave seven behind)

\_\_\_\_\_

13.  $127 \div 30$  — the remainder

\_\_\_\_\_

14. \$100 shared by 8 — dollars each, rounded down

\_\_\_\_\_

15. 95 eggs into boxes of 12 — full boxes

\_\_\_\_\_

16. 95 eggs into boxes of 12 — eggs left over

\_\_\_\_\_

17.  $247 \div 8$  — the remainder

\_\_\_\_\_

18.  $512 \div 15$  — the remainder

\_\_\_\_\_

19.  $1,000 \div 7$  — the remainder

\_\_\_\_\_

20.  $365 \div 12$  — the whole part

\_\_\_\_\_

21.  $365 \div 12$  — the remainder

\_\_\_\_\_

**★ Challenge** NOT GRADED • REASON IT OUT

22.  $17 \div 4$  as a mixed number — type the fraction part as a/b

23. A rope of 250 cm cut into 40 cm pieces. Whole pieces

24. 53 kids in vans of 8 — vans needed

25. 53 kids in vans of 8 — empty seats in the last van

**✎ Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

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# Quotient Hunt & Road Trip

Enrichment. Estimate out loud before computing.

## ✦ Warm-Up 4 ITEMS • RECALL

1 600 miles over 3 days —  
miles a day

\_\_\_\_\_

2 \$120 for 4 days of food —  
dollars a day

\_\_\_\_\_

3  $48 \div 6$

\_\_\_\_\_

4  $100 \div 4$

\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Short Core today — do all of them, then turn the page over.

5. 1,200 miles at 60 mph — hours driving

\_\_\_\_\_

6. A tank does 400 miles. Tanks needed for 1,000 miles (Two and a half is not enough)

\_\_\_\_\_

7.  $336 \div 6$

\_\_\_\_\_

8.  $4,536 \div 21$

\_\_\_\_\_



# Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 1.1 • How Many Fit

OUT OF 22

Warm-Up • 1-12 • 8 | 3 | 3 | 3 | 3 | 4 | 6 | 6 | 9 | 7 | 8 | 9

Core • 13-22 • 13 | 30 | 21 | 32 | 20 | 12 | 6 | 8 | 17 | 11

Challenge • 23-26 • 50 | 80 | 25 | 20

## 1.2 • Long Division

OUT OF 22

Warm-Up • 1-12 • 7 | 6 | 5 | 7 | 12 | 12 | 12 | 21 | 32 | 14 | 15 | 30

Core • 13-22 • 23 | 28 | 27 | 34 | 46 | 56 | 54 | 85 | 92 | 124

Challenge • 23-26 • 132 | 112 | 147 | 297

## 1.3 • Four Digits

OUT OF 22

Warm-Up • 1-12 • 100 | 200 | 300 | 210 | 200 | 210 | 100 | 100 | 600 | 600 | 300 | 500

Core • 13-22 • 141 | 216 | 203 | 184 | 378 | 306 | 392 | 294 | 169 | 396

Challenge • 23-26 • 369 | 255 | 312 | 249

## 1.4 • Remainders

OUT OF 21

Warm-Up • 1-11 • 2 | 3 | 3 | 1 | 1 | 3 | 1 | 2 | 6 | 3 | 6

Core • 12-21 • 5 | 7 | 12 | 7 | 11 | 7 | 2 | 6 | 30 | 5

Challenge • 22-25 • 1/4 | 6 | 7 | 3

## Fri • Quotient Hunt & Road Trip

OUT OF 8

Warm-Up • 1-4 • 200 | 30 | 8 | 25

Core • 5-8 • 20 | 3 | 56 | 216

Challenge • 9-14 • 120 | 30 | 100 | 369 | 19 | 150

# Four Digits by Two

Where the first digit of the quotient goes, and why.

## Warm-Up 12 ITEMS • RECALL

1  $1200 \div 12$

2  $2400 \div 12$

3  $3600 \div 12$

4  $1000 \div 10$

5  $4200 \div 20$

6  $5000 \div 25$

7  $2400 \div 24$

8  $3600 \div 18$

9  $4800 \div 12$

10  $2000 \div 20$

11  $6300 \div 30$

12  $7500 \div 25$

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $4536 \div 21$

14.  $8464 \div 46$

15.  $9072 \div 24$

16.  $3384 \div 24$

17.  $6120 \div 24$

18.  $5,376 \div 28$

19.  $7,616 \div 34$

20.  $8,208 \div 27$

21.  $4,158 \div 22$

22.  $7,290 \div 45$



# Zeros in the Quotient

The place that gets skipped is the place people lose.

## ✦ Warm-Up 12 ITEMS • RECALL

1  $600 \div 6$

2  $800 \div 8$

3  $606 \div 6$

4  $404 \div 4$

5  $900 \div 9$

6  $1000 \div 5$

7  $700 \div 7$

8  $909 \div 9$

9  $505 \div 5$

10  $800 \div 4$

11  $1200 \div 6$

12  $606 \div 3$

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $7308 \div 36$

14.  $4080 \div 8$

15.  $6036 \div 6$

16.  $2050 \div 25$

17.  $5025 \div 25$

18.  $8,120 \div 40$

19.  $6,090 \div 6$

20.  $5,040 \div 5$

21.  $3,045 \div 15$

22.  $7,035 \div 35$



# Estimate the Quotient

A rough answer means a wrong one gets noticed.

## Warm-Up 12 ITEMS • RECALL

1 Estimate  $400 \div 20$ 

\_\_\_\_\_

2 Estimate  $600 \div 30$ 

\_\_\_\_\_

3 Estimate  $900 \div 30$ 

\_\_\_\_\_

4 Estimate  $1200 \div 40$ 

\_\_\_\_\_

5 Estimate  $800 \div 20$ 

\_\_\_\_\_

6 Estimate  $1000 \div 50$ 

\_\_\_\_\_

7 Estimate  $500 \div 25$ 

\_\_\_\_\_

8 Estimate  $800 \div 40$ 

\_\_\_\_\_

9 Estimate  $1500 \div 50$ 

\_\_\_\_\_

10 Estimate  $2400 \div 60$ 

\_\_\_\_\_

11 Estimate  $700 \div 70$ 

\_\_\_\_\_

12 Estimate  $3000 \div 30$ 

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimate  $4850 \div 48$  to the nearest ten

.....

14. Estimate  $3120 \div 39$ 

.....

15. Estimate  $6300 \div 68$  to the nearest ten

.....

16. Estimate  $2450 \div 51$ 

.....

17. Estimate  $7800 \div 19$  to the nearest hundred

.....

18. Estimate  $5,950 \div 59$  to the nearest ten

.....

19. Estimate  $4,180 \div 38$ 

.....

20. Estimate  $7,200 \div 81$  to the nearest ten

.....

21. Estimate  $3,550 \div 71$ 

.....

22. Estimate  $9,900 \div 22$ 

.....



# Divisors Near a Ten

Rounding 29 to 30 makes the guessing quick.

## Warm-Up 11 ITEMS • RECALL

1  $600 \div 30$

2  $620 \div 20$

3  $900 \div 30$

4  $840 \div 40$

5  $960 \div 30$

6  $720 \div 20$

7  $810 \div 27$  — estimate with  $800 \div 27$

8  $1200 \div 40$

9  $640 \div 20$

10  $880 \div 40$

11  $750 \div 30$

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12.  $812 \div 29$

13.  $1178 \div 31$

14.  $1449 \div 21$

15.  $2451 \div 43$

16.  $1764 \div 42$

17.  $957 \div 29$

18.  $1,271 \div 31$

19.  $1,978 \div 46$

20.  $2,632 \div 47$

21.  $1,872 \div 39$



# Road Trip Budget Begins

Pick the route, gather the real numbers.

NAME \_\_\_\_\_

DATE \_\_\_\_\_

RIGHT \_\_\_\_ OF \_\_\_\_ TRIED

✦ Warm-Up

4 ITEMS • RECALL

1 600 miles over 3 days — miles a day

2 \$120 over 4 days — dollars a day

3 900 miles over 3 days

4 \$150 over 5 days

◆ Core

GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. 1200 miles at 60 mph — hours
6. A tank does 400 miles — tanks for 1000 miles
7. 1,500 miles at 60 mph — hours
8. A tank does 350 miles — tanks for 1,050 miles



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Mark the reasoning, not the handwriting.

# Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

## 2.1 • Four Digits by Two

OUT OF 22

Warm-Up • 1-12 • 100 | 200 | 300 | 100 | 210 | 200 | 100 | 200 | 400 | 100 | 210 | 300

Core • 13-22 • 216 | 184 | 378 | 141 | 255 | 192 | 224 | 304 | 189 | 162

Challenge • 23-26 • 369 | 112 | 221 | 174

## 2.2 • Zeros in the Quotient

OUT OF 22

Warm-Up • 1-12 • 100 | 100 | 101 | 101 | 100 | 200 | 100 | 101 | 101 | 200 | 200 | 202

Core • 13-22 • 203 | 510 | 1006 | 82 | 201 | 203 | 1015 | 1008 | 203 | 201

Challenge • 23-26 • 201 | 252 | 251 | 252

## 2.3 • Estimate the Quotient

OUT OF 22

Warm-Up • 1-12 • 20 | 20 | 30 | 30 | 40 | 20 | 20 | 20 | 30 | 40 | 10 | 100

Core • 13-22 • 100 | 80 | 90 | 50 | 400 | 100 | 110 | 90 | 50 | 450

Challenge • 23-26 • 80 | 97 | 110 | 119

## 2.4 • Divisors Near a Ten

OUT OF 21

Warm-Up • 1-11 • 20 | 31 | 30 | 21 | 32 | 36 | 30 | 30 | 32 | 22 | 25

Core • 12-21 • 28 | 38 | 69 | 57 | 42 | 33 | 41 | 43 | 56 | 48

Challenge • 22-25 • 68 | 72 | 101 | 94

## Fri • Road Trip Budget Begins

OUT OF 8

Warm-Up • 1-4 • 200 | 30 | 300 | 30

Core • 5-8 • 20 | 3 | 25 | 3

Challenge • 9-14 • 120 | 30 | 44 | 120 | 40 | 28

# Find the Remainder

It is always smaller than the divisor. Always.

## ✦ Warm-Up 10 ITEMS • RECALL

1  $17 \div 5$  — the remainder

\_\_\_\_\_

2  $23 \div 4$  — the remainder

\_\_\_\_\_

3  $50 \div 6$  — the remainder

\_\_\_\_\_

4  $100 \div 3$  — the remainder

\_\_\_\_\_

5  $20 \div 5$  — the remainder

\_\_\_\_\_

6  $19 \div 6$  — the remainder

\_\_\_\_\_

7  $30 \div 7$  — the remainder

\_\_\_\_\_

8  $45 \div 6$  — the remainder

\_\_\_\_\_

9  $19 \div 2$  — the remainder

\_\_\_\_\_

10  $50 \div 8$  — the remainder

\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

11.  $100 \div 7$  — the remainder

\_\_\_\_\_

12.  $500 \div 23$  — the remainder

\_\_\_\_\_

13.  $750 \div 24$  — the remainder

\_\_\_\_\_

14.  $1000 \div 31$  — the remainder

\_\_\_\_\_

15.  $250 \div 12$  — the remainder

\_\_\_\_\_

16.  $347 \div 9$  — the remainder

\_\_\_\_\_

17.  $1,234 \div 11$  — the remainder

\_\_\_\_\_

18.  $852 \div 25$  — the remainder

\_\_\_\_\_

19.  $999 \div 14$  — the remainder

\_\_\_\_\_

20.  $600 \div 13$  — the remainder

\_\_\_\_\_



NOT GRADED • REASON IT OUT



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

# Round Up or Down

Buses round up. Full boxes round down.

## Warm-Up 12 ITEMS • RECALL

1 100 people, buses hold 30 — buses

\_\_\_\_\_

2 100 pencils, boxes of 30 — full boxes

\_\_\_\_\_

3 50 people, cars hold 4 — cars

\_\_\_\_\_

4 50 pencils, packs of 4 — full packs

\_\_\_\_\_

5 10 people, tables of 4 — tables

\_\_\_\_\_

6 10 pencils, packs of 4 — full packs

\_\_\_\_\_

7 26 kids in vans of 6 — vans needed

\_\_\_\_\_

8 26 kids, whole full vans

\_\_\_\_\_

9 \$50 buys how many at \$8

\_\_\_\_\_

10  $50 \div 8$  — the whole part

\_\_\_\_\_

11 17 seats in rows of 5 — rows needed

\_\_\_\_\_

12  $17 \div 5$  — the whole part

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 200 people, buses hold 45 — buses

\_\_\_\_\_

14. 200 items, boxes of 45 — full boxes

\_\_\_\_\_

15. 1,000 miles at 200 a day — days

\_\_\_\_\_

16. 1,050 miles at 200 a day — days

\_\_\_\_\_

17. 90 cupcakes, trays of 12 — trays

\_\_\_\_\_

18. 100 people in buses of 24 — buses needed

\_\_\_\_\_

19.  $100 \div 24$  — the whole part

\_\_\_\_\_

20. \$250 buys how many at \$30

\_\_\_\_\_

21. 365 days in weeks — whole weeks

\_\_\_\_\_

22. 1,000 books in boxes of 36 — boxes needed

\_\_\_\_\_



## Working space



## Error journal

# Keep It as a Fraction

Pizza does not round. Neither does money.

## Warm-Up 12 ITEMS • RECALL

1 \$10 split 4 ways

\_\_\_\_\_

2 \$5 split 2 ways

\_\_\_\_\_

3 \$9 split 2 ways

\_\_\_\_\_

4 \$20 split 8 ways

\_\_\_\_\_

5 \$6 split 4 ways

\_\_\_\_\_

6 \$12 split 8 ways

\_\_\_\_\_

7  $7 \div 2$  — type as a decimal

\_\_\_\_\_

8  $9 \div 2$ 

\_\_\_\_\_

9  $5 \div 4$ 

\_\_\_\_\_

10  $3 \div 4$ 

\_\_\_\_\_

11  $1 \div 2$ 

\_\_\_\_\_

12  $7 \div 4$ 

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. \$50 split 4 ways, to the cent

.....

14. \$100 split 30 ways, to the cent

.....

15. \$40 split 3 ways, to the cent

.....

16. 7 pizzas split 4 ways, as a decimal

.....

17. \$15 split 4 ways

.....

18.  $13 \div 4$ 

.....

19.  $17 \div 8$ 

.....

20.  $22 \div 5$ 

.....

21.  $9 \div 8$ 

.....

22.  $33 \div 4$ 

.....



## Working space



## Error journal

# Four Questions, One Division

100 ÷ 30 answered four ways, all correct.

## Warm-Up 12 ITEMS • RECALL

1 100 ÷ 30 — the whole part  
\_\_\_\_\_

2 100 ÷ 30 — the remainder  
\_\_\_\_\_

3 100 ÷ 30 rounded up  
\_\_\_\_\_

4 100 ÷ 30 rounded down  
\_\_\_\_\_

5 30 × 3  
\_\_\_\_\_

6 100 − 90  
\_\_\_\_\_

7 25 ÷ 4 — the whole part  
\_\_\_\_\_

8 25 ÷ 4 — the remainder  
\_\_\_\_\_

9 25 ÷ 4 rounded up  
\_\_\_\_\_

10 25 ÷ 4 as a decimal  
\_\_\_\_\_

11 30 ÷ 4 — the whole part  
\_\_\_\_\_

12 30 ÷ 4 as a decimal  
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 100 people, buses of 30 — buses  
.....

14. 100 pencils, full boxes of 30  
.....

15. \$100 split 30 ways, to the cent  
.....

16.  $30 \times 3 + 10$   
.....

17.  $53 \div 8$  — the whole part  
.....

18.  $53 \div 8$  rounded up  
.....

19.  $53 \div 8$  as a decimal  
.....

20.  $53 \div 8$  — the remainder  
.....

21. Which answer suits 53 kids into 8-seat vans — type 7  
.....



## Working space



## Error journal

# Budget the Miles

Miles per day, tanks per trip, dollars per person.

## Warm-Up 8 ITEMS • RECALL

1 600 miles over 3 days — miles a day

2 \$120 over 4 days — dollars a day

3  $1200 \div 6$

4  $400 \times 3$

5  $60 \times 20$

6  $240 \times 5$

7 \$600 over 6 days

8 900 miles over 3 days

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

9. 1,200 miles at 60 mph — hours

10. A tank does 400 miles — tanks for 1,200

11. \$1,440 over 12 days

12. 960 miles over 4 days

13. 2,400 miles at 55 mph — hours, rounded up

14. \$1,450 over 10 days

15. 2,100 miles over 7 days



## SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

# Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 3.1 • Find the Remainder

OUT OF 20

Warm-Up • 1-10 • 2 | 3 | 2 | 1 | 0 | 1 | 2 | 3 | 1 | 2

Core • 11-20 • 2 | 17 | 6 | 8 | 10 | 5 | 2 | 2 | 5 | 2

Challenge • 21-24 • 23 | 23 | 10 | smaller

## 3.2 • Round Up or Down

OUT OF 22

Warm-Up • 1-12 • 4 | 3 | 13 | 12 | 3 | 2 | 5 | 4 | 6 | 6 | 4 | 3

Core • 13-22 • 5 | 4 | 5 | 6 | 8 | 5 | 4 | 8 | 52 | 28

Challenge • 23-26 • 6 | 5 | 18 | 19

## 3.3 • Keep It as a Fraction

OUT OF 22

Warm-Up • 1-12 • 2.5 | 2.5 | 4.5 | 2.5 | 1.5 | 1.5 | 3.5 | 4.5 | 1.25 | 0.75 | 0.5 | 1.75

Core • 13-22 • 12.5 | 3.33 | 13.33 | 1.75 | 3.75 | 3.25 | 2.125 | 4.4 | 1.125 | 8.25

Challenge • 23-26 • 14.29 | 10 | 12.5 | 33.33

## 3.4 • Four Questions, One Division

OUT OF 21

Warm-Up • 1-12 • 3 | 10 | 4 | 3 | 90 | 10 | 6 | 1 | 7 | 6.25 | 7 | 7.5

Core • 13-21 • 4 | 3 | 3.33 | 100 | 6 | 7 | 6.625 | 5 | 7

Challenge • 22-25 • decimal | up | 6.63 | 6

## Fri • Budget the Miles

OUT OF 15

Warm-Up • 1-8 • 200 | 30 | 200 | 1200 | 1200 | 1200 | 100 | 300

Core • 9-15 • 20 | 3 | 120 | 240 | 44 | 145 | 300

Challenge • 16-20 • 240 | 6 | 166.67 | 17 | 300

# Finish the Budget

Every line item divided out and totalled.

## Warm-Up 12 ITEMS • RECALL

1 3 days at \$120

\_\_\_\_\_

2 \$360 + \$200

\_\_\_\_\_

3  $5 \times \$40$ 

\_\_\_\_\_

4 \$600 – \$560

\_\_\_\_\_

5  $4 \times \$50$ 

\_\_\_\_\_

6 \$1,000 ÷ 5

\_\_\_\_\_

7 \$1,200 over 10 days

\_\_\_\_\_

8 \$900 over 6 days

\_\_\_\_\_

9 1,500 miles over 5 days

\_\_\_\_\_

10 \$60 a day for 8 days

\_\_\_\_\_

11  $\$40 \times 12$ 

\_\_\_\_\_

12  $2,000 \div 4$ 

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 12 days at \$120

\_\_\_\_\_

14. 3 tanks at \$65

\_\_\_\_\_

15.  $\$1,440 + \$195$ 

\_\_\_\_\_

16. \$1,635 split 4 people, to the cent

\_\_\_\_\_

17. 1,200 miles over 3 tanks

\_\_\_\_\_

18. \$1,850 over 12 days — to the cent

\_\_\_\_\_

19. 2,750 miles over 11 days

\_\_\_\_\_

20. \$95 a night for 9 nights

\_\_\_\_\_

21. \$2,400 less \$855 lodging

\_\_\_\_\_

22. \$1,545 over 12 days — to the cent

\_\_\_\_\_



## Working space



## Error journal

# Defend the Rounding

Pick three lines and say why each rounded the way it did.

## Warm-Up 12 ITEMS • RECALL

1 2.9 tanks of fuel — tanks bought  
\_\_\_\_\_

2 2.9 rounded down  
\_\_\_\_\_

3 Round 13.334 to the cent  
\_\_\_\_\_

4 Round 2.5 up  
\_\_\_\_\_

5 Round 2.4 down  
\_\_\_\_\_

6 Round \$9.996 to the cent  
\_\_\_\_\_

7 Round 148 to the nearest ten  
\_\_\_\_\_

8 Round 1,480 to the nearest hundred  
\_\_\_\_\_

9 Round \$12.49 to the nearest dollar  
\_\_\_\_\_

10 Round 6.7 hours to the nearest hour  
\_\_\_\_\_

11 Round 249 to the nearest hundred  
\_\_\_\_\_

12 Round 55 to the nearest ten  
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 1,250 miles, a tank does 400 — tanks needed  
.....

14. 1,250 miles — full tanks used  
.....

15. \$50 split 3 ways, to the cent  
.....

16. Cents left over on that split  
.....

17.  $1,200 \div 400$   
.....

18. 18.4 days of driving — days to plan for  
.....

19. \$154.17 a day rounded for planning — to the nearest ten dollars  
.....

20. 98.2 gallons — gallons to buy  
.....

21. For fuel, round the estimate — type up or down  
.....

22. For money left over, round — type down or up  
.....

★

Challenge

NOT GRADED • REASON IT OUT

23. 1,250 miles with 4 tanks bought — miles of spare range
- .....
24. \$1,635 split 12 ways, to the cent
- .....
25. A budget rounded down each day by \$4.17 over 12 days — the shortfall
- .....
26. Rounding a 18.4-day trip down to 18 days — miles left undriven at 150 a day
- .....



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

# Mixed Review

Long division and remainder interpretation together.

## Warm-Up 12 ITEMS • RECALL

1  $84 \div 12$

2  $372 \div 3$

3  $100 \div 7$  — the whole part

4  $100 \div 7$  — the remainder

5  $600 \div 30$

6  $936 \div 24$

7  $48 \div 6$

8  $100 \div 4$

9  $13 \div 4$  — the remainder

10  $7 \div 2$  as a decimal

11  $900 \div 30$

12  $606 \div 3$

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $4,536 \div 21$

14.  $7,308 \div 36$

15.  $812 \div 29$

16.  $8,464 \div 46$

17.  $750 \div 24$  — the remainder

18.  $336 \div 6$

19.  $957 \div 29$

20.  $365 \div 12$  — the remainder



# Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

## Warm-Up 11 ITEMS • RECALL

1  $72 \div 8$

2  $618 \div 6$

3  $63 \div 9$

4  $300 \div 5$

5  $120 \div 4$

6  $360 \div 6$

7  $144 \div 12$

8  $20 \div 8$  as a decimal

9  $31 \div 7$  — the remainder

10  $800 \div 40$

11  $505 \div 5$

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12.  $936 \div 24$

13.  $3,120 \div 39$

14.  $1,178 \div 31$

15.  $9,072 \div 24$

16.  $2,451 \div 43$

17.  $432 \div 8$

18.  $5,376 \div 28$

19.  $8,120 \div 40$

20.  $1,271 \div 31$

21. 100 people in buses of 24 — buses needed



# Mission 03 Test

Twelve items plus the Big Question, answered out loud.

## Warm-Up 4 ITEMS • RECALL

1 618 ÷ 6

2 884 ÷ 34

3 56 ÷ 7

4 120 ÷ 4

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** This one is a test. Do **every** problem — no stopping early.

5. 8,464 ÷ 46

6. 9,072 ÷ 24

7. 6,036 ÷ 6

8. 1,178 ÷ 31

9. 750 ÷ 24 — the remainder

10. Estimate 6,300 ÷ 68 to the nearest ten

11. 100 people, buses of 30 — buses

12. 100 pencils, full boxes of 30

13. 595 ÷ 7

14. 4,158 ÷ 22

15. 6,090 ÷ 6

16. 1,978 ÷ 46

17. Estimate 4,180 ÷ 38



## Working space



## Error journal

# Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 4.1 • Finish the Budget

OUT OF 22

Warm-Up • 1-12 • 360 | 560 | 200 | 40 | 200 | 200 | 120 | 150 | 300 | 480 | 480 | 500

Core • 13-22 • 1440 | 195 | 1635 | 408.75 | 400 | 154.17 | 250 | 855 | 1545 | 128.75

Challenge • 23-26 • 136.25 | 365 | 99 | 356.4

## 4.2 • Defend the Rounding

OUT OF 22

Warm-Up • 1-12 • 3 | 2 | 13.33 | 3 | 2 | 10 | 150 | 1500 | 12 | 7 | 200 | 60

Core • 13-22 • 4 | 3 | 16.67 | 1 | 3 | 19 | 150 | 99 | up | down

Challenge • 23-26 • 350 | 136.25 | 50.04 | 60

## 4.3 • Mixed Review

OUT OF 20

Warm-Up • 1-12 • 7 | 124 | 14 | 2 | 20 | 39 | 8 | 25 | 1 | 3.5 | 30 | 202

Core • 13-20 • 216 | 203 | 28 | 184 | 6 | 56 | 33 | 5

Challenge • 21-24 • 201 | 4 | 369 | 33.33

## Thu • Error Journal Sweep

OUT OF 21

Warm-Up • 1-11 • 9 | 103 | 7 | 60 | 30 | 60 | 12 | 2.5 | 3 | 20 | 101

Core • 12-21 • 39 | 80 | 38 | 378 | 57 | 54 | 192 | 203 | 41 | 5

Challenge • 22-25 • 1006 | 17 | 249 | 6.625

## Fri • Mission 03 Test

OUT OF 20

Warm-Up • 1-4 • 103 | 26 | 8 | 30

Core • 5-20 • 184 | 378 | 1006 | 38 | 6 | 90 | 4 | 3 | 85 | 189 | 1015 | 43 | 110 | 5 | 2.125 | 120

Challenge • 21-24 • 3.33 | 6 | 312 | 19